Qns	Answer	Marks
3a	Let speed of X before hitting Y be v_i Gain in KE = Loss in GPE $\frac{1}{2}mv_i^2 = mgh_i$ $v_i = \sqrt{2gh_i} = \sqrt{2\times 9.81\times 1.50\sin 40°}$ $v_i = 4.35 \text{ m s}^{-1}$	 **B0** **M0** **A1**
3bi	Let speed of X after hitting Y be v_f Loss in KE = Gain in GPE $\frac{1}{2}mv_f^2 = mgh_f$ $v_f = \sqrt{2gh_f} = \sqrt{2\times 9.81\times 0.80\sin 40°}$ $v_f = 3.18 \text{ m s}^{-1}$ **EITHER** Since collision is elastic, Relative speed of approach = relative speed of separation $v_i = v_y - v_f$ 4.35 = v_y - (- 3.18) v_y = 1.17 m s^{-1} **OR** By conservation of linear momentum, $m_X u_X = m_X v_X + m_Y v_Y$ $30(4.35+3.18) = m_Y v_Y \quad \cdots(1)$ By conservation of energy Loss in KE of X = Gain in KE of Y $\frac{1}{2}m_X u_X^2 - \frac{1}{2}m_X v_X^2 = \frac{1}{2}m_Y v_Y^2$ $30(4.35^2 - 3.18^2) = m_Y v_Y^2 \quad \cdots(2)$ Taking (2) ÷ (1) $v_Y = 1.17 \text{ m s}^{-1}$	 **M1** **M1** **A1** (either eqn correctly written) **M1** **A1**

Qns	Answer	Marks
3bii	Gain in KE of Y = Loss in GPE of X $\frac{1}{2}m_Y v_y{}^2 = m_X g\Delta h$ $m_Y = \frac{2m_X g\Delta h}{v_y{}^2}$ $m_Y = \frac{2(0.030)(9.81)\sin 40°(1.50-0.80))}{1.17^2}$ $m_Y = 0.192$ kg $m_Y = 190$ g **OR** From (b)(i) Using the law of conservation of momentum in (1) and energy in (2) $30^2(4.35+3.18)^2 = m_Y{}^2 v_Y{}^2 \quad \cdots(1)^2$ $30(4.35^2-3.18^2) = m_Y v_Y{}^2 \quad \cdots(2)$ Taking $(1)^2 \div (2)$ $m_Y = 193$ g = 190 g (2 s.f.)	B1 M1 A0 B1 M1 A0
3ci	The collision is inelastic because the <u>relative speed of approach</u> is non-zero but the <u>relative speed of separation is zero</u>, which means that they are <u>not equal</u>.	A1
3cii	By conservation of linear momentum, $m_X v_i = (m_X + m_Y)v'$ $v' = \frac{m_X v_i}{m_X + m_Y}$ $v' = \frac{(0.030)(4.35)}{0.030+0.190}$ $v' = 0.5932$ m s^{-1} $v' = 0.59$ m s^{-1}	M1 A0

Qns	Answer	Marks
3ciii	Loss in KE = Gain in GPE + work done against friction $\frac{1}{2}(m_X + m_Y)(v')^2 = (m_X + m_Y)gd\sin 50° + fd$ $\frac{1}{2}(m_X + m_Y)(v')^2 = ((m_X + m_Y)g\sin 50° + f)d$ $d = \dfrac{\frac{1}{2}(0.030 + 0.190)(0.59^2)}{(0.030 + 0.190) \times 9.81 \times \sin 50° + 2.2}$ $d = 0.0099$ m OR Resultant force along the slope = friction + component of weight $ma = 2.2 + mg\sin 50°$ $a = \dfrac{2.2}{0.220} + (9.81)\sin 50°$ $a = 17.51 \text{ m s}^{-2}$ where a is directed downslope. Using $v^2 = u^2 + 2as$, where s is the distance travelled up the slope, d. $0 = (0.59)^2 - 2(17.51)d$ $d = 0.0099$ m	B1 M1 A1 (M1) (C1) (A1)

Qns	Answer	Marks
4a	Electric force per unit positive charge	